

C2-09 Meridian Senior / BI Capstone

Independent integrated project - no solved answer

COMPANY + EXECUTIVE REQUEST

Meridian Commerce & Services is synthetic. Leadership sees revenue growth but worries about growth quality: uneven gross margin, potentially weak West acquisition efficiency, deteriorating customer health in some segments and increasing service pressure. Decision: Where is growth healthy, where is it fragile, and what should leadership do next quarter?

SOURCE PACKAGE

sales_lines.csv = invoice-line Sales fact; customers.csv/products.csv = dimension candidates; marketing_spend.csv = Month-Region-Channel fact; regional_targets.csv = Month-Region fact; customer_health.json = Customer-Month snapshot fact; support_tickets.csv = Ticket fact with deliberate duplicate/missing-priority traps; source_contracts/control_totals + optional PostgreSQL schema.

NON-NEGOTIABLE

Do not flatten all facts; state grain before joins; preserve raw sources; do not call suggestive patterns causal proof; no solved SQL/PBIX/Python/executive answer is supplied.

Meridian capstone deliverables

CAP1-CAP14 evidence chain

BUILD CHAIN

CAP1 Decision contract -> CAP2 Source inventory -> CAP3 SQL extraction -> CAP4 governed transformation -> CAP5 semantic model -> CAP6 investigation -> CAP7 decision report -> CAP8 independent validation -> CAP9 executive brief -> CAP10 handover -> CAP11 technical defense -> CAP12 interview stories -> CAP13 promotion case -> CAP14 job-switch/production reflection.

MINIMUM TECHNICAL EVIDENCE

Maintainable SQL or precise executable SQL design; transformed CSV/JSON evidence; semantic model/PBIX or equivalent detailed model specification; measures/report evidence; at least 10 validation controls including supplied raw controls and quality traps.

ENVIRONMENT BOUNDARY

PostgreSQL and Power BI are strongly recommended where available. Live Service/Fabric execution is optional if unavailable; record ENVIRONMENT BLOCKED rather than fabricating production proof.

Raw controls and quality traps

Independent truth anchors from the frozen data package

RAW CONTROL TOTALS

Sales rows 576; Net Revenue 1,923,183; Cost 868,250; Gross Margin 1,054,933; Customers 60; Marketing rows 216 / Spend 13,068,135; Targets 72 / total 16,758,000; CustomerHealth objects 1,080; Support rows 271 / unique ticket IDs 270.

DELIBERATE TRAPS

Support contains exactly one duplicated ticket row and one blank Priority. Marketing/Target/Health/Support grains differ from Sales invoice-line grain. West spend and later West/Mid-Market health patterns are investigation evidence only.

RELEASE RULE

If a critical raw/grain/key/reconciliation/definition/security/integrity check fails, do not release the executive recommendation. A beautiful report cannot override failed controls.

Capstone submission package

What you hand in before opening protected review

REQUIRED PACKAGE

Decision/metric contract; source/grain profile; SQL/extracts; transformation log; model/measure contracts; analysis O/I/R register; report evidence; validation log; executive brief; README/dictionaries/lineage/runbook/known issues; technical defense; interview stories; promotion/job-switch evidence links.

TRUTH + PRIVACY

Synthetic results must be labeled as project evidence. Do not claim employer revenue, users, scale, salary impact or production operation. If you later add workplace evidence, sanitize names/data/credentials/proprietary artifacts before any portfolio/interview use.

AFTER SUBMISSION

Open the protected CAP review criterion page(s), preserve the first attempt, repair only failed competencies and complete changed retries. After CAP14 mastery, move to Final Gate A.